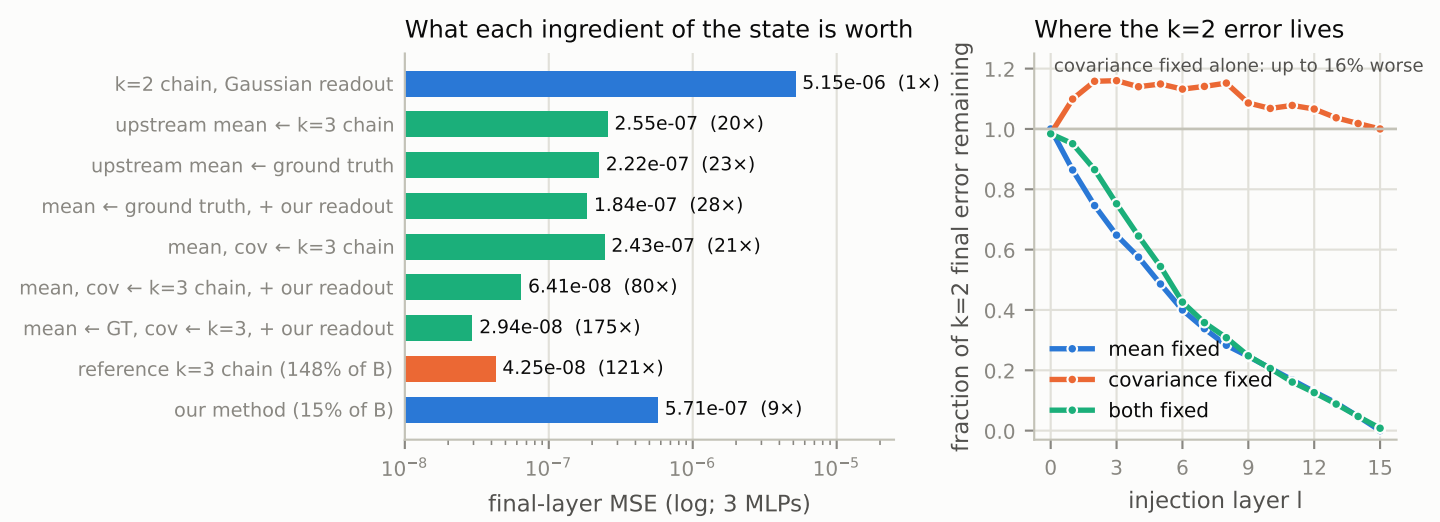
What each ingredient of the state is worth
k=2 chain, Gaussian readout
upstream mean ← k=3 chain
upstream mean ← ground truth
mean ← ground truth, + our readout
mean, cov ← k=3 chain
mean, cov ← k=3 chain, + our readout
mean ← GT, cov ← k=3, + our readout
reference k=3 chain (148% of B)
our method (15% of B)
5.15e-06 (1×)
2.55e-07 (20×)
2.22e-07 (23×)
1.84e-07 (28×)
2.43e-07 (21×)
6.41e-08 (80×)
2.94e-08 (175×)
4.25e-08 (121×)
5.71e-07 (9×)
10⁻⁸
10⁻⁷
10⁻⁶
10⁻⁵
final-layer MSE (log; 3 MLPs)
Where the k=2 error lives
covariance fixed alone: up to 16% worse
fraction of k=2 final error remaining
1.2
1.0
0.8
0.6
0.4
0.2
0.0
mean fixed
covariance fixed
both fixed
0
3
6
9
12
15
injection layer l
deployable (computed by us end to end)
ARC reference k=3 (over budget)
oracle-injected at layers 0–14